

# Traditional Risk Factors versus Novel Cardiometabolic Indices in Predicting High-Risk Plaque and Coronary Calcification in Diabetes Mellitus

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## Abstract

**Background:** Cardiovascular disease is a major complication of Diabetes Mellitus (DM). Computed Tomography Coronary Angiography (CTCA) is widely used to identify High-Risk Plaque (HRP) and coronary artery calcification (CAC). In addition to conventional risk factors, various novel cardiometabolic indices have been proposed; however, their real-world clinical application remains debated and limited, particularly among the diabetic population.

**Objective:** To evaluate the association of traditional clinical risk factors (age, gender, chronic kidney disease [CKD], hypertension, smoking status), laboratory parameters (triglycerides [TG], high-density lipoprotein [HDL], low-density lipoprotein [LDL], total cholesterol, serum uric acid [SUA], HbA1c), and novel cardiometabolic indices (TyHbA1c Index, TG/HDL Ratio, Non-HDL Cholesterol, SUA/HDL Ratio) with the presence of HRP and CAC (Agatston score).

**Methods:** A cross-sectional study was conducted on DM patients undergoing CTCA at RSUP Ngoerah, Denpasar, in 2025. Data were analyzed using bivariate comparative tests and internally validated multivariate logistic regression to evaluate the independent contribution of each clinical and metabolic factor.

**Results:** A total of 52 DM patients with complete records were analyzed. Sixteen patients (30.7%) were newly diagnosed with DM during the initial evaluation prior to CTCA. Among the 36 patients with a known history of DM, 17 (32.7%) were on oral antidiabetic medications, 17 (32.7%) were on insulin regimens, and only 2 (3.8%) received no DM pharmacological therapy. Most patients were referred by cardiologists due to a prior history of cardiac events, and 37 (71.2%) were already on statin therapy. Based on CTCA imaging, the proportion of HRP was high at 71.2%. Bivariate analysis demonstrated that the HRP-positive group was significantly older than the HRP-negative group (mean age 63.16 vs. 55.8 years). Absolute calcification scores also showed a significant difference (median 556.7 vs. 11.6 Agatston,  $p < 0.001$ ). None of the derivative cardiometabolic indices (TyHbA1c Index, TG/HDL Ratio, Non-HDL Cholesterol, and SUA/HDL Ratio) showed statistically significant differences between the groups in this limited cohort ( $p > 0.05$ ). In multivariate modeling, age emerged as the sole statistically significant independent predictor for both HRP formation (aOR 1.08;  $p = 0.045$ ) and  $CAC \geq 100$  (aOR 1.10;  $p = 0.021$ ). CKD Stage  $\geq 3$  indicated a clinical trend of increased HRP risk but did not reach statistical significance due to the small sample size.

**Conclusion:** Aging is the primary determinant driving plaque vulnerability and calcification burden in DM patients. Novel cardiometabolic indices do not provide additional predictive value, likely due to the secondary stabilizing effects of high-intensity statin therapy received by this cohort prior to CTCA.

**Keywords:**

Cardiometabolic Index; Coronary Artery Calcification; Diabetes Mellitus; High-Risk Plaque.

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